

Assignment 4

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Problem (Munkres 23.10). Let $\{X_\alpha\}_{\alpha \in J}$ be an indexed family of connected spaces; let X be the product space

$$X = \prod_{\alpha \in J} X_\alpha.$$

Let $a = (a_\alpha)$ be a fixed point of X .

- (a) Given any finite subset K of J , let X_K denote the subspace of X consisting of all points $x = (x_\alpha)$ such that $x_\alpha = a_\alpha$ for $\alpha \notin K$. Show that X_K is connected.
- (b) Show that the union Y of the spaces X_K is connected.
- (c) Show that X equals the closure of Y . Conclude that X is connected.

Proof of (a). $K \subseteq J$ is a finite subset, say

$$K = \{\alpha_1, \dots, \alpha_n\}.$$

Then $x = (x_\alpha) \in X_K$ coincides with a on all, but finitely many x_α , namely the indices

$$\{x_{\alpha_i}\}_{i \in [n]}.$$

More importantly, $x, y \in X_K$ coincide on all but these finitely many indices.

The most straightforward approach for connectedness is to show that

$$X_K \cong \prod_{i \in [n]} X_{\alpha_i}.$$

Connectedness is preserved under homeomorphism.

Let $\varphi : X_K \rightarrow \prod_{i \in [n]} X_{\alpha_i}$ be the map that sends

$$(x_\alpha) \rightarrow (x_{\alpha_1}, x_{\alpha_2}, \dots, x_{\alpha_n}).$$

Since, for all $x, y \in X_K$, they coincide on all but these finitely many indices.

Thus there is a natural inverse $\varphi^{-1} : \prod_{i \in [n]} X_{\alpha_i} \rightarrow X_K$ which sends $(x_{\alpha_1}, \dots, x_{\alpha_n})$ to the $x = (x_\alpha) \in X_K$ for which the finitely many nonfixed indices are $x_{\alpha_1}, x_{\alpha_2}, \dots, x_{\alpha_n}$.

Then φ^{-1} is the function mapping

$$(x_{\alpha_1}, \dots, x_{\alpha_n}) \mapsto y = (y_\alpha)_{\alpha \in J}$$

where $y_{\alpha_i} = x_{\alpha_i}$ and $y_\alpha = a_\alpha$ otherwise.

This map is well-defined since $y = \varphi^{-1}(x_{\alpha_1}, \dots, x_{\alpha_n})$ coincides with a on all, but the finitely many indices.

It is not terribly hard to see that $\varphi^{-1} \circ \varphi \equiv \text{id}_{X_K}$ and that $\varphi \circ \varphi^{-1} \equiv \text{id}_{\prod X_\alpha}$.

It remains to prove that these maps are continuous.

For the continuity of φ , consider a basis element

$$\prod_{i=1}^n U_i$$

of $\prod_{i=1}^n X_{\alpha_i}$, where each U_i is open in X_{α_i} . Then

$$\varphi^{-1} \left(\prod_{i=1}^n U_i \right)$$

consists of those points of X_K whose α_i -coordinate lies in U_i for each $i \in [n]$. Thus

$$\varphi^{-1} \left(\prod_{i=1}^n U_i \right) = X_K \cap \prod_{\alpha \in J} U_\alpha,$$

where

$$U_{\alpha_i} = U_i$$

for $i \in [n]$, and

$$U_\alpha = X_\alpha$$

otherwise. This is open in X_K by the subspace topology. Hence φ is continuous.

For the continuity of φ^{-1} , consider a basis element

$$X_K \cap \prod_{\alpha \in J} U_\alpha$$

of X_K , where each U_α is open in X_α and $U_\alpha = X_\alpha$ for all but finitely many α .

If $a_\alpha \notin U_\alpha$ for some $\alpha \notin K$, then

$$(\varphi^{-1})^{-1} \left(X_K \cap \prod_{\alpha \in J} U_\alpha \right) = \emptyset,$$

which is open.

Otherwise, $a_\alpha \in U_\alpha$ for every $\alpha \notin K$, and

$$(\varphi^{-1})^{-1} \left(X_K \cap \prod_{\alpha \in J} U_\alpha \right) = \prod_{i=1}^n U_{\alpha_i},$$

which is open in $\prod_{i=1}^n X_{\alpha_i}$.

Therefore φ^{-1} is continuous. Hence

$$X_K \cong \prod_{i=1}^n X_{\alpha_i}.$$

Since each X_{α_i} is connected and a finite product of connected spaces is connected,

$$\prod_{i=1}^n X_{\alpha_i}$$

is connected. Therefore X_K is connected. □

Proof of (b). For each finite $K \subseteq J$, let

$$Y = \bigcup_{K \subseteq J} X_K.$$

Let $f: Y \rightarrow \{0,1\}$ be any continuous function.

We will work to show that this function is necessarily constant.

For $K \subseteq J$, let f_K denote the restriction $f|_{X_K}$.

Then, since X_K is connected, f_K must necessarily be constant.

We suppose, for a contradiction, that f is nonconstant.

Then, necessarily, there are some $K, K' \subseteq J$ such that

$$\text{im } f_K \neq \text{im } f_{K'}.$$

Suppose, without loss of generality, that $\text{im } f_K = 0$ and $\text{im } f_{K'} = 1$.

Every X_K contains the fixed point a , since $a_\alpha = a_\alpha$ for every coordinate outside K .

So

$$a \in X_K \cap X_{K'}.$$

Then

$$f(a) = f_K(a) \neq f_{K'}(a) = f(a),$$

a contradiction.

Thus Y must be connected. □

Proof of (c). Take an arbitrary point $x = (x_\alpha) \in X$.

We want to show that every open neighborhood of x intersects Y .

Consider a basic open neighborhood

$$U = \prod_{\alpha \in J} U_\alpha$$

of x , where $U_\alpha = X_\alpha$ for all but finitely many α .

Let

$$K = \{\alpha \in J \mid U_\alpha \neq X_\alpha\}.$$

$K \subseteq J$ is finite.

Let $y \in X_K \subseteq Y$ coincide with x on all indices $\alpha \in K$.

Then, for $\alpha \in K$ we have

$$y_\alpha = x_\alpha$$

while for $\alpha \notin K$,

$$U_\alpha = X_\alpha$$

so $y_\alpha = a_\alpha \in U_\alpha$.

Thus

$$U \cap Y \neq \emptyset.$$

Since x and its neighborhood were arbitrary, every point of X lies in $\overline{Y}$.

Thus

$$\overline{Y} = X.$$

Since the closure of a connected subspace is connected, we have that

$$X = \overline{Y}$$

is connected. □